How Many Documents Cover a Corpus

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Abstract

A sample drawn from a survey population comes with two numbers: how far its estimates may sit from the population's, and how many more units a target precision needs. A sample drawn from a corpus to annotate or to summarize comes with neither. We supply both for one question, how much of the corpus lies within a chosen semantic distance r of some sampled document. The share that does not, the missing mass at resolution r, has an estimator G that needs no labels and that counts the sampled documents with no sampled neighbour within r (Maurer, 2022). We show that G as a function of sample size n and resolution r collapses onto one curve in nr^d, where d is the intrinsic dimension of the embedded corpus estimated from a 2,000-document pilot, so d is the corpus property that sets the sample size, in the role the variance plays in a survey formula. A fit on samples of at most 1,000 documents predicts the isolation rate at 7,000 documents to a mean error of 0.03 to 0.05 on three real corpora. For a finite corpus of N documents the missing mass is $G\,(1 - n/N)$, the finite population correction of survey sampling, to a mean error of 0.01. The collapse fails on a corpus whose local dimension runs from 2 to 16 across its clusters, so the fit's R^2 diagnoses whether one number describes a corpus. A second question, how many documents are needed to see every theme of a corpus, is a question about a partition, and its answer follows from the theme masses alone: for thirty themes it is the same curve in every one of eight corpora. Ten corpora from three platforms, two encoders and three seeds support these results.

1 Introduction

Large collections of text are sampled before they are studied. An annotation team codes a few thousand posts out of millions, a language model summarizes a few hundred documents because the corpus does not fit its context, and a topic model is fitted to a slice because the full run is slow. The sampler of a human population has a settled answer to the question such a slice raises. Given the population variance and a target error, the formula $n = z^2\sigma^2/e^2$ gives the sample size, and the estimate comes with a margin of error (Cochran, 1977; Kish, 1965). The sampler of a corpus has no such formula. The sample size is set by budget, and whether the slice covers the corpus is left unasked.

The question has an exact form once a corpus is a set of points. Embed each document, choose a distance r that says what counts as the same content, and ask what share of the corpus lies farther than r from every sampled document. That share is the missing mass at resolution r. In the discrete case, where two documents either share a category or do not, it is the quantity Good and Turing estimated from the number of categories seen once (Good, 1953), and ecologists have built a sampling theory on it, with rarefaction and extrapolation curves that say how many more units a target coverage needs (Chao and Jost, 2012). Maurer (2022) extended the estimator to metric spaces. His G is the share of sampled points with no other sampled point within r, and he proved its bias and variance bounds. The estimator has had no application to text, and nothing in it says how G falls as the sample grows.

This paper turns G into a sample size formula for corpora and tests it on ten corpora. Four results carry the paper.

1. The isolation rate $G(n, r)$ depends on n and r only through nr^d, where d is the intrinsic dimension of the embedded corpus. A single curve fitted on samples of at most 1,000 documents predicts the isolation rate at 7,000 documents with a mean error of 0.03 to 0.05 on AskDocs, TIFU and a tweet corpus, and the fit recovers d on synthetic manifolds of known dimension. The required sample size for a target isolation rate at resolution r then

follows in closed form, and d, estimated from a pilot of 2,000 documents, is the corpus property that sets it.

2. For a finite corpus of N documents the missing mass is $G\,(1-n/N)$. The correction is the one survey sampling applies to a finite population, and it brings the error of G against the true missing mass from 0.07 to 0.01.

3. The collapse fails on one corpus of political posts whose clusters have local dimension from 2 to 16, slogans beside discussion, and it keeps failing after near duplicates are removed. The R^2 of the collapse fit is therefore a diagnostic that says whether one dimension describes a corpus.

4. How many documents are needed to see every theme is a different question. For a fixed number of themes its answer depends on the theme masses alone, and it is the same curve in all eight corpora we tested, 0.82 to 0.86 of the theme mass at $n = 50$. Coverage at a fixed resolution differs tenfold across the same corpora. The two questions need two tools, and we say which answers which.

In practice the procedure is short. A user picks the resolution r that matches the question, embeds a pilot, estimates d, and takes the sample size and the coverage band. If the collapse fit is poor the corpus has more than one scale, and the user is told so before the sample is drawn.

2 Background

2.1 Sample coverage in the discrete case

A sample of n draws from a population of categories covers the categories it has seen. The mass it has missed is the total probability of the unseen categories, and Good (1953) showed that the share of draws whose category appeared exactly once, f_1/n, estimates it with small bias. Ecologists took the estimator as the basis of a sampling theory for species. Chao (1984) bounded the number of unseen species from the singletons and doubletons f_1 and f_2, and Chao and Jost (2012) wrote the expected coverage of a sample enlarged by m draws as a closed form in n, f_1 and f_2, so that a survey of a habitat can be stopped at a target coverage instead of a target size. Efron and Thisted (1976) asked how many words Shakespeare knew with the same machinery, and Orlitsky et al. (2016) gave the optimal prediction of the number of unseen species up to a sample about $n \log n$ times larger than the one in hand.

The estimators have reached text and machine learning a few times. Hajič and Moss (2025) apply the Chao bound to a catalogue of chant to say when a collection is close to complete. Tran et al. (2017) fit species models to the answers of an open ended survey question to predict when new answers stop appearing. Chung et al. (2018) count unknown training examples with the same estimator, with a species defined as a unique example, and Pal et al. (2026) estimate the mass of deployment states a classifier has never seen. In each of these a species is a discrete label, two items either match or do not, and the question of how many more items a target needs is answered only by Hajič and Moss (2025) and Tran et al. (2017).

2.2 The missing mass in a metric space

Maurer (2022) removed the discreteness. In a metric space the missing mass at scale r of a sample $X_1, \dots, X_n$ is the probability that a new draw lies farther than r from every X_i, and the estimator becomes

$$G(X, r) = \frac{1}{n} \left| \{k : d(X_k, X_i) > r \text{ for all } i \neq k\} \right|, \tag{1}$$

the share of sample points with no other sample point within r. In the discrete case, with $d(x, y) = 1$ for $x \neq y$ and $r < 1$, it is f_1/n. Maurer proved that G overestimates the conditional missing mass by at most $1/n$ in expectation and that the variance of the difference is at most $3/n$ (his Theorem 2.1). The paper is theory: it sketches anomaly detection and coding as applications, and it contains no data, no embeddings and no statement about how G falls with n. The nearest ecological construction is the attribute diversity of Chao et al. (2019), in which two species closer than a threshold τ count as partly the same and diversity is reported as a profile in τ. That profile is the analogue of our treatment of r as a resolution the user chooses.

2.3 Diversity of a sample in embedding space

Several measures describe how spread out a set of embedded points is. The Vendi score is the exponential of the entropy of the kernel matrix's eigenvalues, an effective number of distinct items (Friedman and Dieng, 2023), and its cousins extend it to the orders of a Hill number (Pasarkar and Dieng, 2024). Precision and recall for generative

models count points inside the nearest neighbour balls of a reference set (Kynkäänniemi et al., 2019), density and coverage correct their bias (Naeem et al., 2020), and MAUVE compares two distributions on a quantized embedding space (Pillutla et al., 2021). Each describes the sample in hand or compares two samples. None asks what a larger sample would cover, and none has a parameter in the role of a population variance.

2.4 Saturation in practice

Where a sample size for text is set at all, it is set by a stopping rule. Qualitative researchers stop interviewing when new interviews add no new code, and a review of 23 empirical tests puts that point at 9 to 17 interviews (Hennink and Kaiser, 2022; Guest et al., 2006). De Paoli and Mathis (2024) carry the rule to coding by a language model and track the slope of the count of unique codes. TopicGPT reports that its topic count levels off after about 600 documents and stops when 200 documents add no topic (Pham et al., 2024). Maeda et al. (2025) fit an model with eight topics to 522 open ended responses and find that 100 responses recover over 60% of the full corpus's topics and 250 recover about 70%. These are observations of a coverage curve. The present paper gives the curve a parameter and an estimator.

3 Method

3.1 Setting and estimand

A corpus is a finite set of N documents. Each is embedded by a fixed encoder and normalised, and the distance between two documents is the cosine distance of their embeddings. A sample is n documents drawn without replacement. The user chooses a resolution r: two documents within r of each other count as the same content for the purpose at hand. The quantity to estimate is the missing mass at resolution r,

$$M(n, r) = \frac{1}{N} \left| \{ j : d(x_j, x_i) > r \text{ for all sampled } i \} \right|, \tag{2}$$

the share of corpus documents with no sampled document within r. Coverage is $1 - M$. The estimator is Maurer's G of Equation 1, computed on the sample alone from each sampled document's distance to its nearest sampled neighbour.

The estimand is the finite corpus in hand. Sampling exists because the corpus is too large to annotate or to place in a model's context, and the full corpus is available to check the estimate. This is the setting of the experiments below, and it is what makes a finite population correction necessary.

3.2 The collapse law

Suppose the embedded documents lie near a manifold of intrinsic dimension d with a smooth density. The mass of a ball of radius r around a document is then of order r^d, and the probability that none of n sampled documents falls in it is of order $\exp(-c\, n\, r^d)$. Averaged over documents, the isolation rate G should then depend on n and r only through the product nr^d. A density that varies across the manifold turns the single exponential into an average of exponentials with different rates, which is slower than exponential and well described by a stretched exponential. We therefore fit

$$\log\big(-\log G(n, r)\big) = a + b \log\big(n\, r^d\big), \tag{3}$$

with d estimated beforehand and a and b fitted by least squares over the cells of an (n, r) grid with $0.02 < G < 0.98$. An exponent $b = 1$ is the pure exponential, and $b < 1$ is the stretched form.

The dimension d comes from the TwoNN estimator of Facco et al. (2017), which uses only the ratio of each point's second to first nearest neighbour distance, on a pilot of 2,000 documents. We also fit Equation 3 with $\log n$ and $\log r$ as separate regressors. The ratio of the $\log r$ coefficient to the $\log n$ coefficient is then an estimate of d that owes nothing to TwoNN, and agreement between the two is a check on the law.

Solving Equation 3 for n gives the sample size at which the isolation rate at resolution r reaches a target ε,

$$n^*(\varepsilon, r) = \left(\frac{\log(1/\varepsilon)}{e^a\, r^d} \right)^{1/b}. \tag{4}$$

This is the corpus analogue of $n = z^2\sigma^2/e^2$. The target ε plays the part of the tolerated error, the resolution r is set by the question, and d is the property of the corpus that the formula needs, in the role of the variance. Each halving of r multiplies n^* by $2^{d/b}$.

3.3 Finite population correction

G estimates the missing mass of the distribution that generated the corpus. When the estimand is the corpus itself, each sampled document is covered at distance zero, so the finite corpus's missing mass is

smaller by the share of the corpus that was sampled. The corrected estimator is

$$\hat{M}(n, r) = G(n, r)\left(1 - n/N\right), \qquad (5)$$

with $1 - n/N$ the finite population correction of survey sampling. The correction is negligible at $n \ll N$ and halves the estimate at $n = N/2$.

3.4 A diagnostic for corpora with more than one scale

Equation 3 assumes one dimension for the whole corpus. The R^2 of the fit, and its R^2 on cells with larger n than those used to fit, say whether the assumption holds. When it fails, two probes locate the cause: the share of documents with a near duplicate (a nearest neighbour within cosine distance 0.02), and the spread of TwoNN dimension across the corpus's k-means clusters.

3.5 Theme coverage as a question about a partition

A different question is how many documents are needed to see every theme. Once themes are a partition of the corpus with masses $m_1, \ldots, m_K$, the expected mass of themes present in a sample of n draws is

$$C_K(n) = \sum_{k=1}^{K} m_k \left(1 - (1 - m_k)^n\right), \qquad (6)$$

the Good-Turing expectation, and it depends on the corpus only through the masses. We test this with inventories of $K = 30$ themes per corpus, with the exact label coverage of each sample and with the themes a language model reports after reading the sample. Section 5.5 shows that this curve and the missing mass at a fixed r behave differently across corpora, which is why the two questions need two tools.

4 Data

Table 1 lists the corpora. Two synthetic families have a known answer. Gaussian manifolds place 20,000 points on a random d-dimensional subspace of $\mathbb{R}^{64}$ for $d \in \{2, 4, 8\}$, so the true intrinsic dimension is known and the collapse law can be checked against it. Zipf topic mixtures draw 20,000 documents from 50 to 500 topic centres with Zipf weights and Gaussian noise, so the topic coverage of any sample is known exactly.

Three forum corpora come from the evaluation of a corpus summarizer (Morris et al., 2023):

Corpus	N	Encoder	d
Manifold d=2	20,000	none	2.0
Manifold d=4	20,000	none	4.2
Manifold d=8	20,000	none	7.6
Zipf, 200 topics	20,000	none	11.0
AskDocs	10,000	GTR	13.4
Trump	10,000	GTR	9.3
TIFU	10,000	GTR	17.7
Census tweets, 5 slices	7,610 to 30,136	Qwen3	7.9 to 13.0
Truth Social	50,000	Qwen3	TBD
Bluesky	50,000	Qwen3	TBD

Table 1: Corpora. d is the TwoNN intrinsic dimension of a 2,000-document pilot (mean over seeds where several exist).

AskDocs (medical questions from Reddit), Trump (posts about the 2020 election), and TIFU (stories told in the first person), 10,000 documents each after cleaning and deduplication, embedded with GTR-T5-base (Ni et al., 2022) at three random seeds. Five corpora are keyword slices of a 2020 export of tweets about the United States Census: random, citizens, trump, covid and illegal, 7,610 to 30,136 tweets after URL and mention removal and deduplication. Two corpora are uniform samples of 50,000 English posts each from Truth Social and from Bluesky between May and December 2024, drawn from full platform collections; the Truth Social collection is conditioned on the 2024 candidates and the Bluesky collection is a broad political pool. These nine corpora are embedded with Qwen3-Embedding-8B (Zhang et al., 2025), 4,096 dimensions, 256 tokens, normalised. Cosine distance is used throughout.

For the theme study each of the eight forum and census corpora received an inventory of 30 themes: k-means on the full Qwen3 embeddings, each cluster named and defined by Gemma 4 (Gemma Team, 2026) from its 15 documents nearest the centroid. The same model summarized or enumerated 1,200 samples, 30 independent draws at each $n \in \{25, 50, 100, 200, 400\}$ per corpus, at temperature 0, with documents truncated to 600 characters.

5 Results

5.1 The estimator tracks true topic coverage

On the Zipf mixtures, where every document's topic is known, coverage $1 - G$ at a radius of half the median pairwise distance of a pilot tracks the true topic mass coverage of the sample with Spearman $\rho = 0.968$ and a mean absolute error of 0.029,

across 50 to 500 topics, two noise levels, three seeds and eleven sample sizes from 50 to 5,000. Good-Turing on the true labels, which the estimator does not see, reaches $\rho = 0.989$ and error 0.016. At smaller radii the agreement falls away ($\rho = 0.31$ at a tenth of the median distance). The radius matters, and the next result removes it as a free choice.

5.2 The isolation rate collapses on nr^d

Table 2 reports the one variable fit of Equation 3 with d fixed at its TwoNN estimate, over an (n, r) grid of eleven sample sizes from 100 to 7,000 and twelve radii from the 5th to the 99th percentile of the pilot's nearest neighbour distances. On the three manifolds the fit explains 95% to 97% of the variance of $\log(-\log G)$, against 4% to 6% for $\log n$ alone and 64% to 73% for $\log r$ alone. On AskDocs, TIFU and the random tweet corpus it explains 95% to 97%. Fitted on cells with $n \leq 1{,}000$ and tested on cells with n up to 7,000, the law keeps R^2 of 0.93 to 0.96, and the isolation rate it predicts at $n = 7{,}000$ has a mean absolute error of 0.044 on AskDocs, 0.029 on TIFU and 0.051 on the tweets. Across seeds 42 to 44 on the forum corpora the out of sample R^2 is 0.950 (sd 0.019) on AskDocs and 0.954 (sd 0.004) on TIFU. Figure 1 shows the surface against nr^d.

The fit with two regressors recovers the dimension without TwoNN. The ratio of the $\log r$ to the $\log n$ coefficient is 2.1, 4.4 and 7.3 on the manifolds of true dimension 2, 4 and 8, and 15.4 and 19.2 on AskDocs and TIFU against TwoNN values of 13.6 and 17.7. The exponent b is below 1 everywhere: 0.60 to 0.75 on the Gaussian manifolds, whose density falls off from the centre, and 0.45 on AskDocs and TIFU. The stretched form is the rule.

5.3 Where the collapse fails, and why

Trump and the Zipf mixture fail the fit (out of sample R^2 0.64 for both, and 0.56 with sd 0.09 for Trump across seeds). In a 5,000-document sample of Trump, 3.4% of documents have a near duplicate within cosine distance 0.02, against 0.0% in AskDocs and TIFU, and the TwoNN dimension of its twenty k-means clusters runs from 2.0 to 16.2 (coefficient of variation 0.52, against 0.15 for AskDocs and 0.10 for TIFU). The clusters of dimension near 2 are slogans and the clusters near 16 are discussion. Removing the 266 near duplicates leaves the out of sample R^2 at 0.74, so the cause is the spread of local dimension and one global d cannot describe both kinds of cluster. The Zipf mixture fails for the related reason of unequal topic mass, with no duplicates. The R^2 of the collapse fit is therefore a diagnostic: a corpus that collapses has one dimension and one sample size formula, and a corpus that does not needs a mixture.

5.4 The finite population correction

Table 3 compares G and $G\,(1 - n/N)$ with the true missing mass M of the finite corpus over four corpora (AskDocs, Trump, TIFU and the Zipf mixture), six sample sizes from 100 to 5,000 and six radii, 144 cells. Uncorrected, G has a mean absolute error of 0.068 and a maximum of 0.42, reached at $n = 5{,}000$ of $N = 10{,}000$, where G is twice M. Corrected, the mean error is 0.015, the maximum 0.11 and the Spearman correlation 0.996. Across the three seeds on the forum corpora the corrected error is 0.010 on AskDocs, 0.013 on TIFU and 0.009 on Trump, so the correction holds on the corpus that fails the collapse as well.

5.5 Theme coverage is a question about a partition

With thirty themes per corpus, the share of theme mass present in a sample is the same in all eight forum and census corpora at every n: 0.58 to 0.65 at $n = 25$, 0.82 to 0.86 at $n = 50$, 0.94 to 0.98 at $n = 100$ and 0.99 to 1.00 at $n = 200$ (Figure 2). Equation 6, evaluated on each corpus's theme masses, reproduces these values with a mean absolute error of 0.005. Over the same samples, coverage at a fixed resolution ($1 - G$ at $r = 0.34$) runs from 0.003 on AskDocs to 0.77 on citizens at $n = 25$. A quantity that is the same everywhere cannot be ordered by one that differs tenfold, and the Spearman correlation across corpora between the two is accordingly near zero at every n.

The language model's enumeration behaves as the partition says it should. Within each corpus the mass of themes Gemma 4 reports present rises with n along the label coverage curve (Spearman $+1.00$ in seven corpora and $+0.90$ in TIFU), and its precision against the label truth rises from 0.73 at $n = 25$ to 0.98 at $n = 400$. Merging the eight inventories into one shared taxonomy at four merge radii (42 to 187 shared themes) opens up differences across corpora, and those differences are predicted by the shared masses through Equation 6 (Spearman $+0.93$ to $+1.00$) and not by G at any fixed radius ($+0.31$, $+0.21$, $+0.38$ and -0.40 at $n = 50$ for the four merges). Appendix C gives the full table.

Corpus	d (TwoNN)	R^2, nr^d	R^2, n only	R^2, r only	out of sample R^2	error at n=7,000
Manifold d=2	2.01	0.970	0.060	0.635	0.952	0.032
Manifold d=4	4.21	0.954	0.044	0.666	0.924	0.056
Manifold d=8	7.60	0.971	0.054	0.731	0.939	0.031
AskDocs	13.6	0.973	0.023	0.801	0.931	0.044
TIFU	17.7	0.972	0.041	0.826	0.956	0.029
Tweets (random)	14.5	0.946	0.033	0.863	0.939	0.051
Trump	9.2	0.811	0.000	0.731	0.636	0.064
Zipf, 200 topics	11.0	0.662	0.003	0.558	0.639	0.141

Table 2: Collapse of the isolation surface. R^2 of $\log(-\log G)$ on $\log(nr^d)$, on $\log n$ alone and on $\log r$ alone over all grid cells with $0.02 < G < 0.98$; out of sample R^2 and the mean absolute error of the predicted G at $n = 7{,}000$ come from a fit on cells with $n \leq 1{,}000$. Seed 42; seed means for the forum corpora are in the text.

n	M (true)	G	$G\,(1 - n/N)$
100	0.729	0.749	0.743
200	0.665	0.684	0.672
500	0.565	0.600	0.574
1,000	0.479	0.534	0.486
2,000	0.378	0.463	0.380
5,000	0.211	0.384	0.212

Table 3: Missing mass of the finite corpus, mean over four corpora and six radii at each n. N is 10,000 for the three forum corpora and 20,000 for the Zipf mixture.

The two questions are therefore different functionals of the same geometry. Coverage of a partition is set by the partition's mass vector. The missing mass at resolution r is set by the local geometry at scale r, and the two coincide only when the partition's cells are balls of radius r, which thematic clusters are not. A user who asks "have I seen every theme" should use Equation 6 with the theme masses, which a pilot inventory supplies. A user who asks "is every document within r of something I have read" should use G, the collapse law and Equation 4.

5.6 Truth Social and Bluesky

[Pending: the two 50,000-post samples are being embedded. This subsection will report their d, collapse R^2 and finite population correction once the run lands.]

6 Discussion

The procedure a user follows has four steps. Choose the resolution r that matches the question: a coarse r for "are the main kinds of post represented", a fine r for "has every specific claim been seen". Embed a pilot of about 2,000 documents and estimate d by TwoNN. Fit Equation 3 on nested subsamples of the pilot and check its R^2. Read n^* from Equation 4 and, after the sample is drawn, report $G\,(1 - n/N)$ as the share of the corpus the sample leaves uncovered. No step needs labels, and the whole runs on a laptop once the embeddings exist.

The survey analogy is exact in two places and instructive in a third. The finite population correction is the same factor for the same reason. The dimension d plays the role of the variance: it is the one property of the population that the sample size formula needs, and like the variance it is estimated from a pilot. The third place is the margin of error. A survey's margin bounds the error of a proportion, and a corpus sample's plain mean already does that job well, to within 0.01 for cluster proportions at $n = 1{,}000$ in our data. The missing mass bounds something else, the mass of content the sample has no example of, which is the error that matters for enumeration: listing the themes, finding the rare claim, checking that a summary had the chance to mention something. The two margins answer different questions, and a report of a corpus sample should carry both.

The theme result explains two failures that an applied reader will have met. A free summary written from a sample is stable across draws at a rate that depends on the summarizer's own granularity, and a theme inventory of fixed size covers at the same rate in every corpus. Both readouts are relative to the corpus, and both are blind to how broad the corpus is at any absolute scale. The breadth that the sample size formula is about lives at a resolution, and only a readout at that resolution can see it.

The collapse diagnostic has a use beyond the formula. A corpus whose isolation surface does not collapse is one where slogans and discussion, or several registers, sit at different scales, and a single sample size will oversample one and undersample the other. The probe of local dimension across clusters locates the registers, and a stratified sample

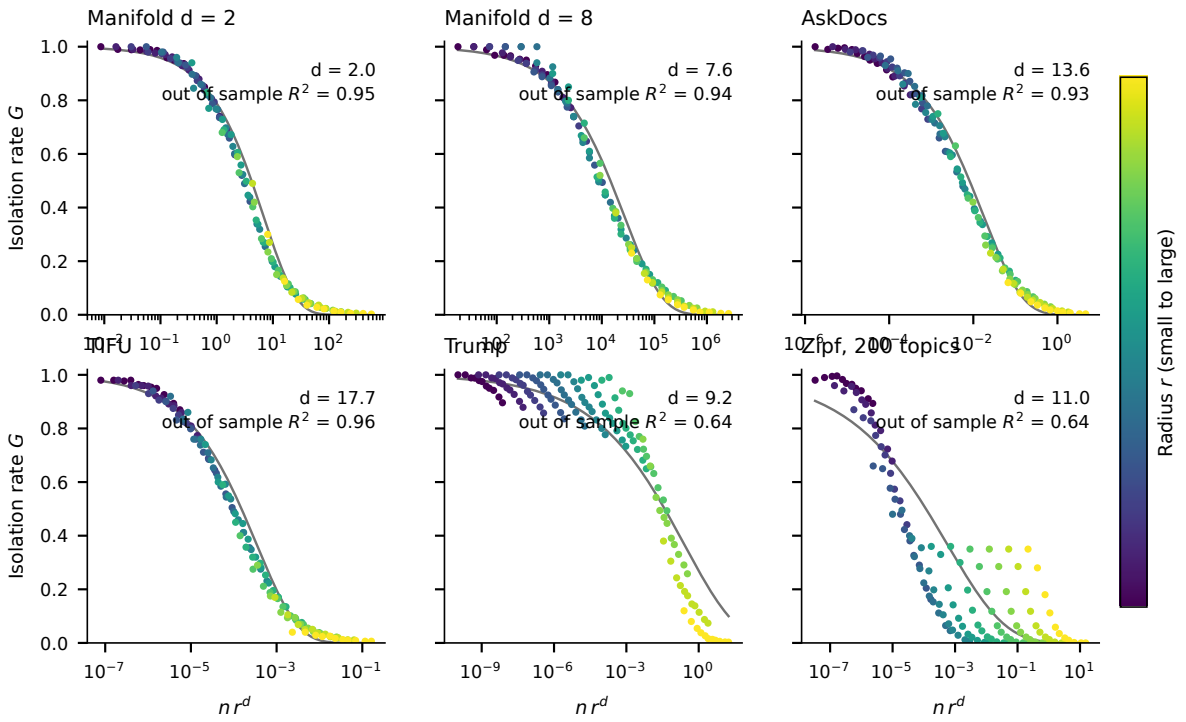


Figure 1: Isolation rate G against nr^d on a log scale, one panel per corpus, colour by radius r. Points from different radii fall on one curve where the law holds (AskDocs, TIFU, the manifolds) and separate where it fails (Trump, the Zipf mixture).

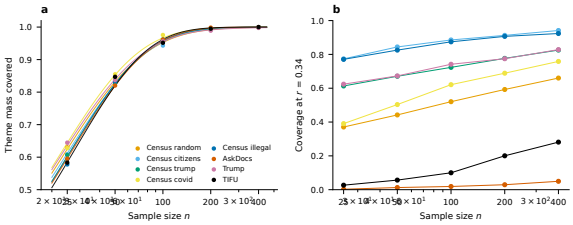


Figure 2: Share of theme mass present in a sample against n, eight corpora with thirty themes each (points), and the Good-Turing expectation of Equation 6 from each corpus's theme masses (lines).

with a size per stratum is the natural next step.

Limitations

The resolution r is a cosine distance in one encoder's space, and the same r means different things in GTR and in Qwen3 embeddings. The collapse law held under both encoders, and d moved with the encoder (13.6 against 12.6 on AskDocs), so a user must fix the encoder before choosing r and must report both.

The exponent b of the stretched exponential is fitted. We have an account of why it is below 1 but no closed form that predicts it from the density, so the formula needs a pilot large enough to fit two coefficients.

The corpora are English social media and forum text from four sources. The Truth Social collection is conditioned on candidate threads, so its d describes that collection and not the platform.

Summary stability was tested and does not follow from coverage across corpora. Within each of eight corpora the instability of summaries written from samples fell with n along the coverage curve, and across corpora at a fixed n the ordering by coverage did not predict the ordering by instability (Appendix B). A summary's stability depends on the scale at which the model abstracts, which this paper does not model.

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A Synthetic smoke test

Zipf topic mixtures with 20,000 documents in 64 dimensions, four settings (50 topics with Zipf exponent 0.5; 200 and 500 topics with exponent 1.0; 200 topics with noise 1.0), three seeds, nested samples at eleven sizes from 50 to 5,000. Coverage $1-G$ at radius α times the median pilot pairwise distance, against the true topic mass coverage: $\alpha = 0.5$ gives $\rho = 0.968$ and error 0.029; $\alpha = 0.4$ gives 0.797 and 0.091; $\alpha = 0.3$ gives 0.677 and 0.209; $\alpha \leq 0.2$ fails. The Chao and Jost extrapolation from a 500-document pilot to the sample size for 95% coverage lands within a factor of two of the empirical crossing in three of four settings and over-predicts on the noisy setting (1,267 against 1,000).

B Summary stability

For each of the eight forum and census corpora and each $n \in \{25, 50, 100, 200, 400\}$, thirty independent samples were summarized by Gemma 4 at temperature 0 in one paragraph of at most 150 words, and the summaries were embedded with Qwen3-Embedding-8B. Instability is the mean pairwise cosine distance among the thirty summaries. Within every corpus instability falls with n and tracks G (Spearman $+0.40$ to $+1.00$, seven of eight at $+0.70$ or above), but the slope of $\log$ instability on $\log G$ runs from $+0.17$ (citizens) to $+24$ (AskDocs). Across corpora at a fixed n, Spearman of instability with G is $+0.14$, $+0.12$, $+0.12$, $+0.07$ and -0.17 at $n = 25$ to 400, with bootstrap bands that include zero, and with d it is $+0.10$ to $+0.29$. The per sample G does not predict how far that sample's summary sits from the centroid (median $\rho = +0.05$ over forty cells). TIFU illustrates the mechanism: its G is near 1 at every shared radius and its summaries are the most stable, because the model summarizes it as stories of mishap whichever thirty stories it is given.

C Shared taxonomy

The eight thirty-theme inventories were merged by average-linkage clustering of their centroids at cosine distance 0.20, 0.15, 0.10 and 0.05, giving 42, 69, 112 and 187 shared themes, and every document was assigned to its nearest shared centroid. Table 4 gives, for each merge and each n, the cross-corpus Spearman of the shared theme coverage of the 1,200 samples with $1-G$ at $r = 0.30$ and with Equation 6 evaluated on the shared masses.

Merge (themes)	n=25	50	100	200	400
with $1-G(n, 0.30)$					
0.20 (42)	+0.26	+0.31	+0.26	+0.29	+0.38
0.15 (69)	+0.24	+0.21	−0.33		
0.10 (112)	+0.88	+0.38	−0.07	−0.24	−0.38
0.05 (187)	+0.26	−0.40	−0.40	−0.40	−0.40
with Equation 6 on the shared masses					
0.20 (42)	+0.98	+1.00	+1.00	+0.98	+0.93
0.15 (69)	+0.93	+0.98	+0.88		
0.10 (112)	+0.93	+0.90	+0.90	+0.98	+0.81
0.05 (187)	+0.98	+0.95	+0.98	+1.00	+1.00

Table 4: Cross-corpus Spearman correlation of shared theme coverage with the two predictors, eight corpora. Blank cells were not computed.

D A bound for functions that are stable at scale r

For a document function f with range 1 that takes the same value on documents within r of each other, the estimate of the corpus mean of f formed by giving each corpus document the value of its nearest sampled document errs by at most $v(r) + M(r)$, where $v(r)$ is the share of covered documents whose nearest sampled document disagrees with them on f. On a family of fifty cluster indicator functions over four corpora, six sample sizes and six radii, the bound held in 144 of 144 cells and was loose for proportions (median 22 times the realised error), where the plain sample mean already errs by 0.003 to 0.07. The bound is tight by construction for presence indicators, where the error equals the missing mass.

E Prompts

Summary. "Below are $\{n\}$ documents sampled from a larger collection. Write a summary of the collection in one paragraph of at most 150 words. Cover the main themes in proportion to how often they appear, name notable minority themes briefly, and do not mention individual document numbers."

Theme naming. "The $\{m\}$ documents below were grouped together because they are about the same theme. Give the theme a short name (at most 8 words) and a one-sentence definition that would let a reader decide whether a new document belongs to it. Answer as JSON with keys name and definition."

Enumeration. "Below are $\{n\}$ documents sampled from a larger collection, followed by a list of $\{K\}$ themes found in the full collection. Decide which themes are present among these $\{n\}$ documents. A theme is present only if you can point to one document that is primarily about it; in a small sample most themes will be absent, and similar-sounding themes are distinct, so pick the single best theme for each document. Answer with one line per present theme in the form theme_id: document_number."